Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_612659835ce4400eaa2fba11ad1399ae
Date	2026-11-01 18:09:42
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
size_mismatch	major	99%	length 118.44 mm (nom 120.00 ±0.10)
size_mismatch	major	99%	hole dia 12.73 (nom 12.00 ±0.05)
size_mismatch	major	99%	hole position off 1.91 mm (±0.10)

2. Root cause

Cause: Improper Bending Force

Reason: The length deviation (118.44mm vs 120.00mm) suggests that excessive force or incorrect settings during the bending process may have caused material deformation beyond the intended dimensions [586f086d-0deb-40c2-a71d-f840aaa55a60].

Effect: Dimensional Inaccuracy

Confidence: 75%

Supporting evidence

- **586f086d-0deb-40c2-a71d-f840aaa55a60**: Improper Bending Force / Over Bending Root Cause Analysis Defect Types: deformation, dimensional inaccuracy Process: sheet metal bending, press brake operations Root Cause: Improper bending force settings or over-bending beyond the material's elastic limit. Effect: Deformation of the part, dimensional inaccuracy in bent features, springback issues causing part to not meet dimensional tolerances. C

3. Solution

Corrective actions (fix this occurrence)

1. Recalibrate press brake force settings to material specification
2. Verify tonnage calculator against actual material hardness test results

Preventive actions (stop it recurring)

1. Implement first-article inspection
2. Standardize tonnage settings for SPCC steel material lots

Precautions

- Ensure material thickness is verified before bending

How to verify the fix

Perform a first-article inspection to confirm length is within 120.00 +/- 0.10mm

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Springback	38%	Elastic recovery of the material after bending can lead to dimensional inaccuracies in linear dimensions related to the bend, which may contribute to the observed length mismatch [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46].
3	Tool Wear / Die Damage	35%	The significant deviation in hole diameter (12.73mm vs 12.00mm) and hole position (1.91mm off) indicates a potential shift or degradation in the punch/die geometry, which is a known effect of tool wear [587aa665-c6f1-48df-8783-09d65ff4f82a].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.